

Overview of mathematical and statistical necessities for population genomics

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Population genomics

Understand genetic processes that shape genomic variation observed in populations.

- ① Develop computational/mathematical models of these population genetic processes.
 - Transmission of genetic material, mutation, recombination, selection, migration, gene-flow, ...
- ② Apply these methods/models to genomic data to learn about the processes.
 - Compute and interpret summary (statistics) of data.
 - Explore processes through simulations.
 - Inference of model parameters from data.

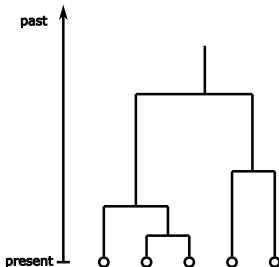
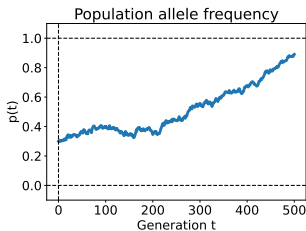
Modelling approaches in population genomics

Forward-in-time:

- If current genetic composition of population is known.
- How will it change into the future?

Backward-in-time (Coalescent theory):

- Observe genomic variation in a population at present.
- Where did it come from? Which ancestors?



Probability and statistics

- Population genetic processes are random.
 - Need models for randomness.
- Statistics needed for analyzing (genomic) data.
 - Analyze genomic data of (randomly) sampled individuals.
 - Handle sampling/measurement uncertainty.

→ Probability and statistics essential for population genomics.

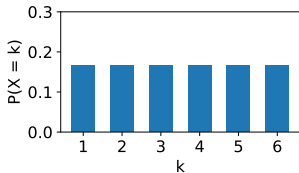
Note

- I will present some background.
- Helpful preparation for the remainder of the course.
- Not possible to cover every detail.
- If you are familiar with most of these topics, that's good.
- If not, possibly read up on some before the course starts.
 - Many accessible textbooks cover these topics.
 - I'd be happy to provide additional material/pointers.
 - Feel free to reach out via email: steinrue@uchicago.edu.
 - Or on slack.

Probability distributions

Example: Throw a die.

- ① Outcomes: 1, 2, ..., 6.
 - $X \in \{1, 2, \dots, 6\}$.
- ② Probabilities: Every number with equal probability.
 - $\mathbb{P}\{X = 1\} = \mathbb{P}\{X = 2\} = \dots = \mathbb{P}\{X = 6\} = \frac{1}{6}$.



<https://en.wikipedia.org/wiki/Dice>

- Often shorthand notation: $p_i := \mathbb{P}\{X = i\} \geq 0$.
- Have to sum to one: $\sum_{\text{all outcomes } k} p_k = 1$.

Expected value

Expected value

If I receive as many Euros as the die shows pips, then how much money should I expect.

- 1 pip w.p. p_1 .
- 2 pips w.p. p_2 .
- ...
- 6 pips w.p. p_6 .

I get k Euro (k pips) with probability p_k .

In general:

$$\mathbb{E}[X] := \sum_{\text{all outcomes } k} k \cdot p_k$$

Specifically for one die: $\mathbb{E}[X] = 3.5$.



<https://en.wikipedia.org/wiki/Dice>

Empirical mean

Unknown distribution of X .

- Estimate the expected value using the empirical mean.
- Repeat random experiment L times, and average outcomes.

Outcome of replicate i : $X^{(i)}$.

Empirical mean

$$\bar{X} := \frac{1}{L} \sum_{i=1}^L X^{(i)}$$

$\bar{X}$ is an estimate of $\mathbb{E}[X]$:

$$\mathbb{E}[X] \approx \bar{X}.$$

Variance

How much does outcome vary around the expected value?

Variance: Expected squared deviation from mean.

Variance

$$\mathbb{V}[X] := \sum_{\text{all outcomes } k} (k - \mathbb{E}[X])^2 \cdot p_k$$

- Small variance: Outcome will be close to expected value.
- Large variance: Large spread of possible values.
- Important when dealing with uncertainty, risk management.

Standard deviation

$$\sigma_X := \sqrt{\mathbb{V}[X]}$$

Common discrete probability distributions

Frequently used basic discrete probability distributions:

- Binomial distribution.
- Geometric distribution.
- Poisson distribution.

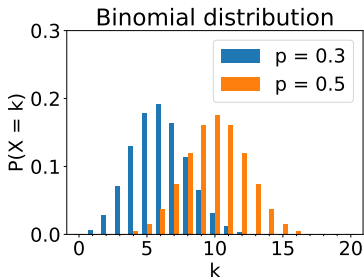
Binomial distribution

- Consider flipping a coin n times.
- Coin has probability p to come up heads.
- Probability $(1 - p)$ to come up tails.

Binomial distribution

Describes the number of times the coin comes up heads.

$$\mathbb{P}\{X = k\} = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$



https://en.wikipedia.org/wiki/Italian_lira

Binomial distribution

$$X \sim \text{Bin}(n, p)$$

$$\mathbb{E}[X] = np \quad \text{and} \quad \mathbb{V}[X] = np(1 - p)$$

Example: Wright-Fisher model

- Offspring inherits alleles from parents.
- Number of alleles in next generation is binomially distributed.

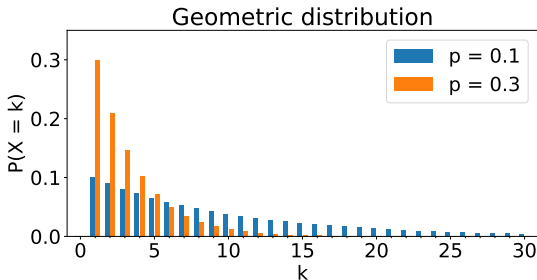
Geometric distribution

- Consider flipping a coin.
- Each coin has probability p to come up heads.

Geometric distribution

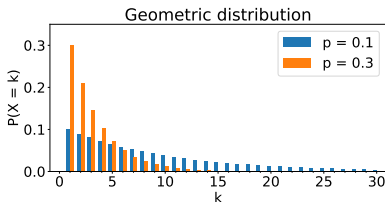
How many times do I have to flip until I see **head** the first time?

$$\mathbb{P}\{X = k\} = (1 - p)^{k-1}p$$



- **Waiting time.**

Geometric distribution



$$X \sim \text{Geom}(p)$$

$$\mathbb{E}[X] = \frac{1}{p} \quad \text{and} \quad \mathbb{V}[X] = \frac{1-p}{p^2}$$

Example: Time to most recent common ancestor

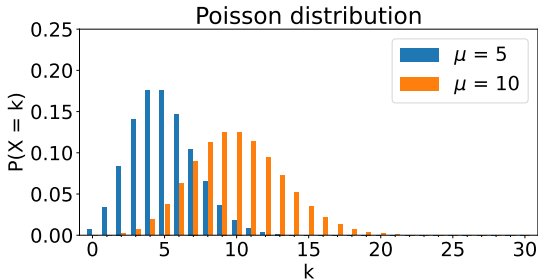
- Two haploids in population of size $2N$.
- Prob. that common parent in previous gener.: $\frac{1}{2N}$.
- # gener. to most recent common ancestor $\sim \text{Geom}(\frac{1}{2N})$.
- Expected # gener. until most recent common ancestor $2N$.

Poisson distribution

- Number of events in a given time interval.
- Expected number of events: μ .

Poisson distribution

$$\mathbb{P}\{X = k\} = \frac{\mu^k e^{-\mu}}{k!}$$



Poisson distribution

$$X \sim \text{Poisson}(\mu)$$

$$\mathbb{E}[X] = \mu \quad \text{and} \quad \mathbb{V}[X] = \mu$$

Example: Number of mutations on evolutionary branches

- Consider gene with $L = 10,000$ bp.
- Per generation per bp mutation prob $u = 10^{-8}$.
- Human-Chimp divergence time $T = 250,000$ generations.
- # mutations in gene between Human-Chimp $\sim \text{Poisson}(\mu)$.
- With $\mu := L \cdot u \cdot 2T = 25$ (expected value).

Continuous probability distributions

We had:

- Discrete probability distributions.
- Used to model discrete outcomes:
 - Number of mutations.
 - Number of generations.
 - Number of alleles.
 - ...

However, some quantities/measurements are on continuous scales:

- Height
- Wingspan
- Cholesterol level
- ...

Need continuous probability distributions.

Continuous probability distributions

Discrete distributions:

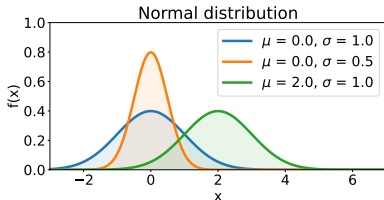
- We defined probability p_k for every outcome k .

Continuous distributions:

- Can't define probability for each outcome (technical reasons).
- Rather, introduce:

Probability density function (pdf): $f(x)$.

- $f(x) \geq 0$ gives likelihood of outcome $x \in \mathbb{R}$.
- Has to integrate to 1: $\int_{-\infty}^{\infty} f(x)dx = 1$.



Expected value & variance

Same interpretation as in discrete case:

- Expected value: Expected outcome.
- Variance: Spread around expected outcome.

However, replace discrete sum with continuous integral:

$$\mathbb{E}[X] := \int_{-\infty}^{\infty} x \cdot f(x) dx$$

and

$$\mathbb{V}[X] := \int_{-\infty}^{\infty} (x - \mathbb{E}[X])^2 \cdot f(x) dx$$

- If limited range, adjust integral boundaries.

Common continuous distributions

Frequently used continuous distributions:

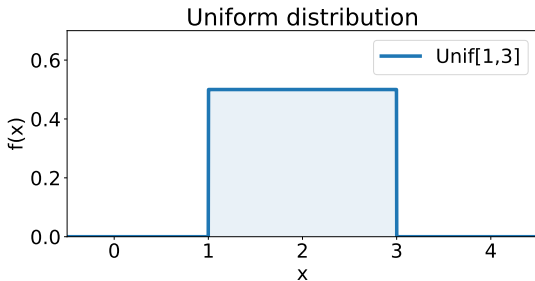
- Uniform.
- Normal distribution.
- Exponential distribution.
- Beta distribution.

Uniform distribution

All values between lower bound a and upper bound b equally likely.

$$X \sim \text{Unif}[a,b]$$

$$f(x) = \begin{cases} \frac{1}{b-a}, & \text{if } a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

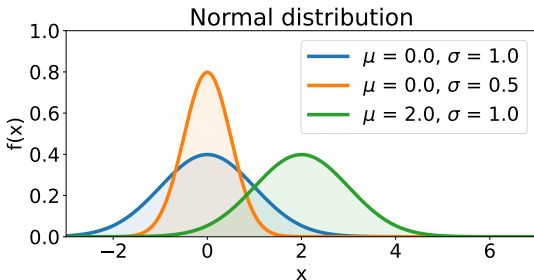


Normal distribution

Very common, flexible distribution:

- Determined by mean μ and standard deviation σ (or variance).

$$X \sim \text{Norm}(\mu, \sigma), \quad f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$



Surprise: $\mathbb{E}[X] = \mu$ and $\mathbb{V}[X] = \sigma^2$

- Sum of many small contributions approx. normal distributed.

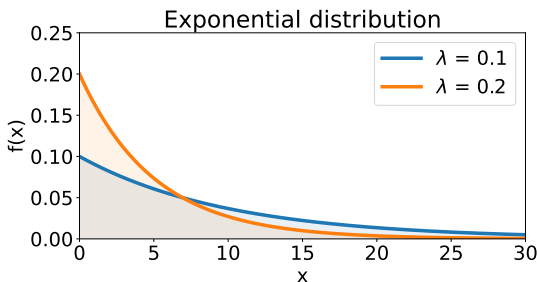
Exponential distribution

Consider an event that happens at rate λ .

Exponential distribution

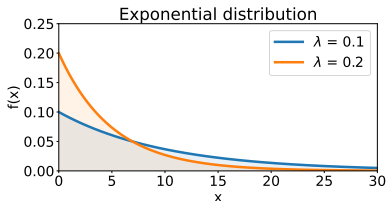
Continuous waiting time until the event happens.

$$X \sim \text{Exp}(\lambda), \quad f(x) = \lambda e^{-\lambda x}$$



$$\mathbb{E}[X] = \frac{1}{\lambda} \quad \text{and} \quad \mathbb{V}[X] = \frac{1}{\lambda^2}$$

Exponential distribution



Example: Time until next mutation.

If mutations happen at rate θ , then time to next mut. $\sim \text{Exp}(\theta)$.

- Waiting time:
 - Continuous time analog of geometric distribution.
- Connection to poisson distribution:
 - If mutations happen at rate θ ,
 - then (# mutations in time interval T) $\sim \text{Poi}(T\theta)$.

Beta distribution

Beta distribution

Flexible probability distribution for random values between 0 and 1.

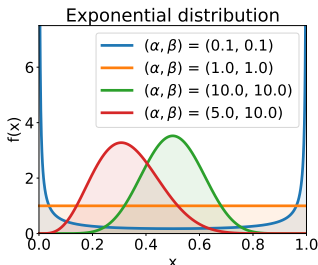
- Two parameters: $\alpha, \beta > 0$.

$$X \sim \text{Beta}(\alpha, \beta)$$

$$f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}$$

$$\mathbb{E}[X] = \frac{\alpha}{\alpha + \beta}$$

- Useful as **prior** for probabilities p , since $p \in [0, 1]$.
- Conjugate prior for binomial distribution.
- Stationary distribution for some Wright-Fisher models.



5 minute intermission.

Population genomic simulations

Some popular tools:

- SLiM (5.0; *Haller, Ralph, Messer; 2025*).
 - Forward-in-time.
 - Very flexible; models various forms of selection.
 - Simulates whole genomes for entire population.
 - msprime (*Baumdicker et al.; 2021*).
 - Backward-in-time (coalescent).
 - Efficient for large scale neutral simulations.
 - Simulates whole genomes for a population sample.
- All build on these elementary distributions.
 - Combine distributions in the right way with the right parameters to model complex population genetic processes.

(Summary) statistics

Compute summary of genomic data.

- Or multiple summaries.

Certain summary (statistics) informative about certain processes:

- Pairwise diversity π :
 - Mutation rate, effective population size.
- F_{ST} , F-statistics:
 - Population structure, selection.
- Tajima's D:
 - Neutrality, selection, effective population size.
- Site-Frequency-Spectrum (SFS):
 - Demography, selection, ...
- ...

Likelihood-based inference of model parameters

Given:

- Observed data D :
 - Genomic sequences of sampled individuals.
 - Alleles of sampled individuals at specific locus.
 - ...
- Model for data D that has parameters $\Theta = (\theta_1, \dots, \theta_n)$:
 - Demographic parameters.
 - Recombination rates.
 - Selection coefficients.
 - ...

Can define

Likelihood: Probability of data given parameter.

$$L(\Theta) := \mathbb{P}_{\Theta}\{D\}$$

If $L(\Theta_1) > L(\Theta_2)$, then data D more likely under parameters Θ_1 .

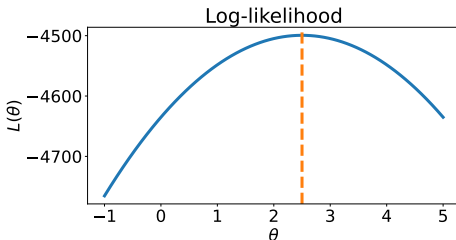
- Parameters Θ_1 explain data better.

Maximum likelihood

Maximum likelihood estimate

Find the parameter Θ^* that maximize the likelihood $L(\Theta)$.

- These parameters explain data D best under given model.
- Equivalently: Maximize log-likelihood: $\log L(\Theta)$.



- Many methods throughout the course infer model parameters from genomic data using this approach.

Next:

Bayesian inference.

But for that we need:

- Conditional probability.
- Bayes' theorem.

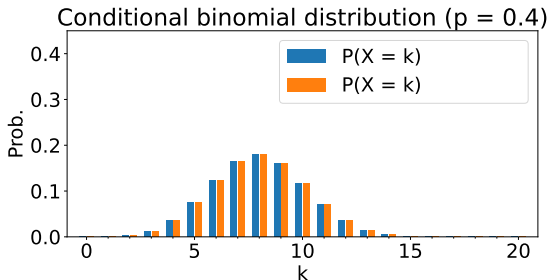
Conditional probability

Probability of specific outcome given information about outcome.

Example: $X \sim \text{Bin}(20, 0.4)$, **conditioned** on $X \geq 8$.

$$\mathbb{P}\{X = k | X \geq 8\} := \frac{\mathbb{P}\{X = k, X \geq 8\}}{\mathbb{P}\{X \geq 8\}}$$

- Restrict distribution, and re-normalize (sum to 1).



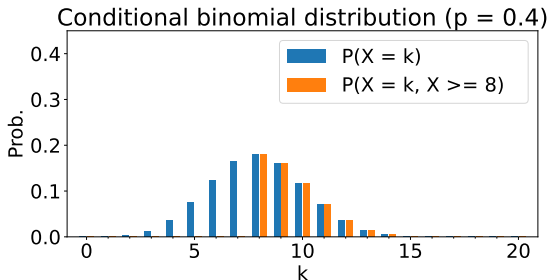
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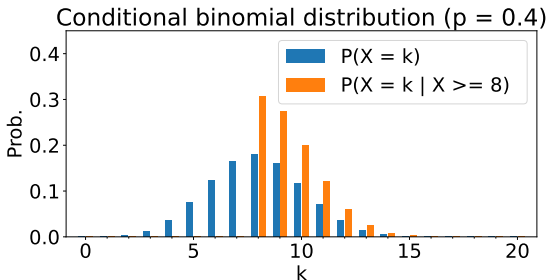
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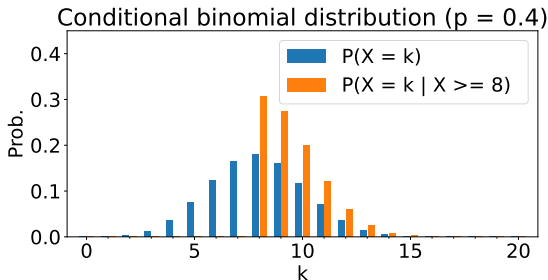
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Conditional probability

In general:

- Probability of outcome A .
- Conditioned on B .

Conditional probability

$$\mathbb{P}\{A|B\} := \frac{\mathbb{P}\{A, B\}}{\mathbb{P}\{B\}}$$

Immediately yields Bayes' theorem:

Bayes' theorem

$$\mathbb{P}\{A|B\} = \frac{\mathbb{P}\{B|A\}\mathbb{P}\{A\}}{\mathbb{P}\{B\}}$$

- Can compute conditional $\mathbb{P}\{A|B\}$ in terms of $\mathbb{P}\{B|A\}$.

Bayesian inference

Interpret:

- A as model parameter Θ .
- B as data D .

Then:

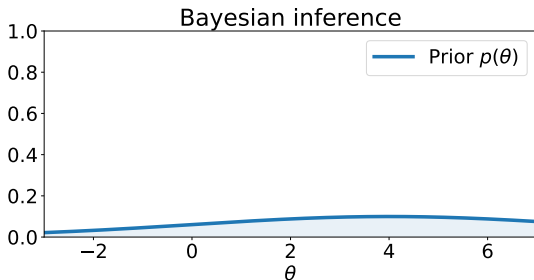
$$\mathbb{P}\{\text{parameter } \Theta | \text{data } D\} = \frac{\mathbb{P}\{\text{data } D | \text{parameter } \Theta\} \mathbb{P}\{\text{parameter } \Theta\}}{\mathbb{P}\{\text{data } D\}}$$

With:

- **Prior:** $\mathbb{P}\{\text{parameter } \Theta\}$
- **Likelihood:** $\mathbb{P}\{\text{data } D | \text{parameter } \Theta\}$
- **Posterior:** $\mathbb{P}\{\text{parameter } \Theta | \text{data } D\}$

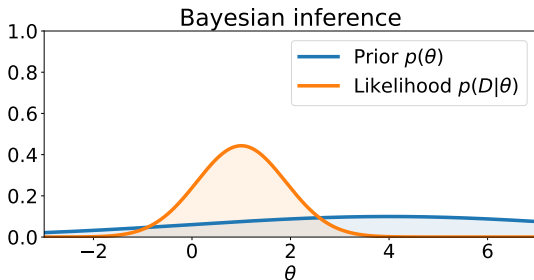
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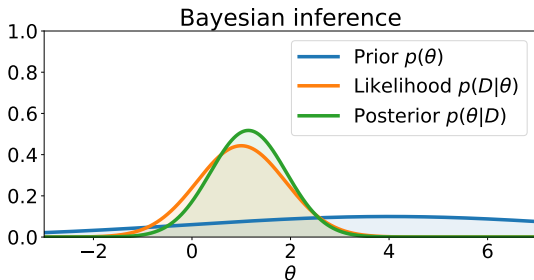
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- Can sometimes be solved analytically.
 - Exact formula for posterior distribution.
- Often analytic solution not possible.
 - Sample from posterior distribution.
 - Monte-Carlo Markov Chain (MCMC).
 - Approximate Bayesian Computation (ABC).
- More details in dedicated lectures.

Latent variables

Latent variable

Random outcome depends on other random outcome.

Example: # mutations sample i and j differ in certain genomic region (of length L bp).

- Per gener. per bp mutation rate u .
- If most recent common ancestor of i and j at T gener. ago.
- Then i and j separated by $2T$ generations.
- # mut. i and j differ is $\sim \text{Poi}(2T \cdot u \cdot L)$.
- But T random, $T \sim \text{Exp}(\frac{1}{2N})$.

$$\begin{aligned}\mathbb{P}\{\pi_{i,j} = k\} &= \int_0^\infty \mathbb{P}\{\pi_{i,j} = k | T\} \mathbb{P}\{T\} dT \\ &= \int_0^\infty \frac{(2T \cdot u \cdot L)^k e^{-k}}{k!} \frac{1}{2N} \exp(-\frac{1}{2N} T) dT\end{aligned}$$

Optimization and gradient descent

Task

Find parameters Θ that maximize/minimize function $L(\Theta)$.

- Maximum-likelihood:
 - Likelihood $L(\Theta)$: How likely observed data for given Θ .
 - Find parameters Θ^* that yield highest likelihood.
 - Explain data best.
- Machine learning:
 - Artificial neural networks (ANNs) have many parameters Θ .
 - Chat-GPT 3.5: 175 billion parameters.
 - Loss function $L(\Theta)$ captures how well ANN performs tasks.
 - Find parameters Θ^* that minimize loss.
 - Performs best on the given tasks.

Often not possible to find global optimum.

- Find Θ where $L(\Theta)$ as large/small as possible.

Maximizing is just minimizing with opposite sign.

Optimization and gradient descent

Minimize $L(\theta)$:

Derivative/gradient

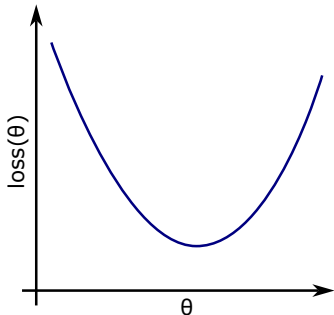
Take derivative of $L(\theta)$ with respect to θ : $\frac{d}{d\theta}L(\theta)$.

- If $\frac{d}{d\theta}L(\theta) > 0$, then $\theta \leftarrow$ yields $L(\theta) \downarrow$.
- If $\frac{d}{d\theta}L(\theta) < 0$, then $\theta \rightarrow$ yields $L(\theta) \downarrow$.

At minimum θ^* :

- Neither $\theta \leftarrow$ nor $\theta \rightarrow$ yields $L(\theta) \downarrow$.
- Thus, $\frac{d}{d\theta}L(\theta^*) = 0$.

In some applications, possible to find θ^* by solving $\frac{d}{d\theta}L(\theta^*) = 0$ analytically.



Optimization and gradient descent

Minimize $L(\theta)$:

Derivative/gradient

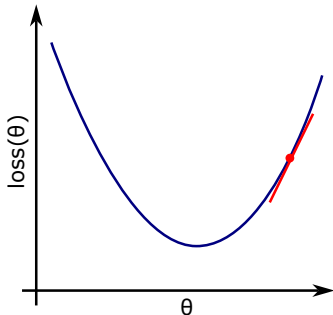
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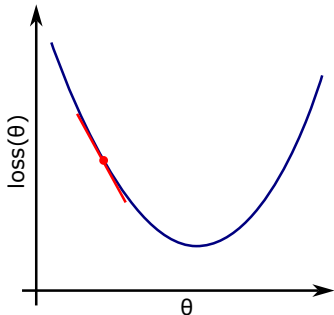
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Optimization and gradient descent

Minimize $L(\theta)$:

Derivative/gradient

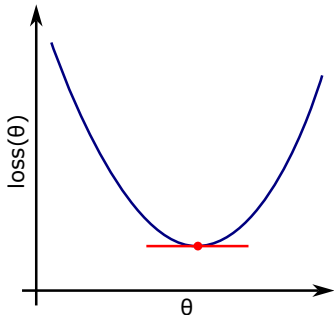
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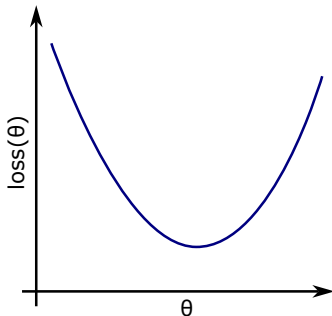


Optimization and gradient descent

If not possible to solve $\frac{d}{d\theta} L(\theta^*) = 0$, then optimize θ stepwise.

Gradient descent

- Start at $\theta^{(0)}$.
- Update $\theta^{(k)} \rightarrow \theta^{(k+1)}$:
 - If $\frac{d}{d\theta} L(\theta^{(k)}) < 0$, then $\theta^{(k)} \rightarrow$.
 - If $\frac{d}{d\theta} L(\theta^{(k)}) > 0$, then $\theta^{(k)} \leftarrow$.
- Until stopping criterion reached.



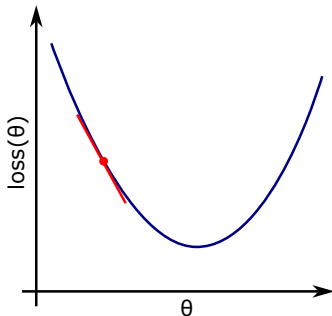
- Many maximum-likelihood methods in population genomics use version of this strategy.
- Almost all training of ANNs uses versions of this.

Optimization and gradient descent

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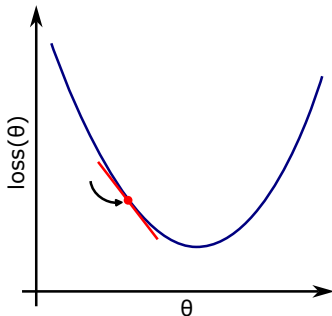
- Many maximum-likelihood methods in population genomics use version of this strategy.
- Almost all training of ANNs uses versions of this.

Optimization and gradient descent

If not possible to solve $\frac{d}{d\theta} L(\theta^*) = 0$, then optimize θ stepwise.

Gradient descent

- Start at $\theta^{(0)}$.
- Update $\theta^{(k)} \rightarrow \theta^{(k+1)}$:
 - If $\frac{d}{d\theta} L(\theta^{(k)}) < 0$, then $\theta^{(k)} \rightarrow$.
 - If $\frac{d}{d\theta} L(\theta^{(k)}) > 0$, then $\theta^{(k)} \leftarrow$.
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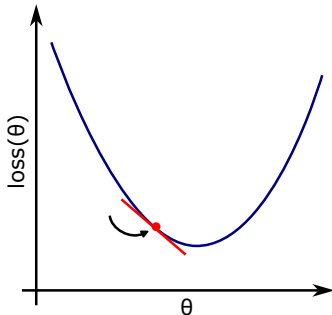
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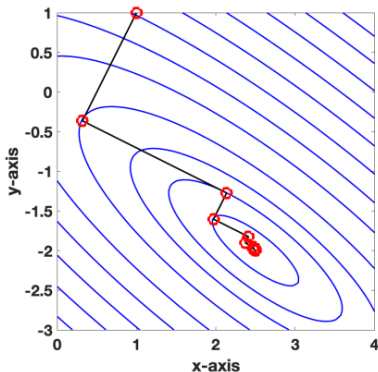


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Optimization and gradient descent

If more than one parameter:

- Gradient descent in multiple dimensions:



https://en.wikipedia.org/wiki/Gradient_descent

That's it for today. Enjoy the course next week!